Talk2MeLatin

Amy Spiteri
Department of Artificial Intelligence, University of Malta
amy.spiteri.23@um.edu.mt

Abstract

Artificial Intelligence has become an increasingly valuable tool in language education, offering interactive learning experiences for widely-studied modern languages. However, there remains a lack of AI-based tools that support learners of classical languages such as Latin, and existing digital resources for Latin do not provide interactive assistance or the ability to interpret learner questions. This project aims to address this gap by developing a conversational AI system capable of understanding grammar-related questions asked by Latin learners and responding with accurate and useful explanations. To achieve this, the project proposes the development of a conversational chatbot system that uses an intent classification model trained to recognise grammar-related user questions, and generates the appropriate responses.

1 Introduction

1.1 Motivation

In recent years, the use of Artificial Intelligence in education, particularly in language learning, has been increasingly growing. The increase in AI-based features in language learning platforms such as Duolingo, and recently the use of large language models, shows the growing recognition of AI's potential to enhance language education. These tools offer many advantages, including personalized learning experiences for the individual student's needs, as well as on-demand access to language support beyond the classroom, making them increasingly valuable in modern educational environments. Conversational interfaces and chatbots, in particular, are effective tools for practising a language as they enable learners to engage in interactive dialogue and provide instant feedback, simulating the benefits of a human tutor, and make learning more accessible for new students. The use of these tools in helping with the study of widely-spoken modern languages demonstrates how useful conversational AI in language education can be.

Despite these advances, classical languages such as Latin do not benefit from the same amount of digital resources and AI-supported educational tools that several modern languages do, creating a significant gap. Latin learners face several unique challenges. Unlike modern languages, Latin is a primarily written and academic language, with no community of native speakers. This means that students cannot rely on conversational immersion or interaction with fluent speakers as language practice, as having someone fluent in Latin available to speak to all of the time may not be possible. Although there are digital resources for Latin, they tend to focus on functionalities such as dictionary lookup or morphological parsing. Tools such as Whitaker's Words (Whitaker, 1993), while valuable for their morphological parsing capabilities, offer only lookup functionality and do not provide interactive explanations or assist learners in a conversational way through natural-language questions. Similarly, traditional Latin dictionaries and grammar references, although reliable, require learners to independently navigate complex grammatical concepts without interactive support.

Although Latin today has no native speakers, it is still studied by many students. It is still relevant in fields such as classical studies, history, law, medicine, linguistics, and theology. The existence of a modern student base with limited tools introduces the need for something that can make Latin more accessible to learners, which can be achieved by expanding the use of AI for this purpose.

Therefore, the motivation of this project is to create a conversational AI tool that helps with Latin grammar as a way to support Latin learners, connecting the gap between classical studies and AI for language learning.

1.2 Problem Definition

Having established the value of AI and conversational tools in language learning and the gap in such resources for Latin, the main problem this project addresses is the absence of a conversational AI system that can understand and respond to grammar related questions about Latin.

As a result, students studying Latin lack a system that can support them when they encounter specific grammatical difficulties, and they do not have a conversational tool to ask for immediate assistance. Existing computational resources for Latin function as dictionaries by providing lookup data, but they lack the interactive capability to answer specific questions.

Therefore, the specific problem this project seeks to solve can be defined as the need for an AI-based system capable of understanding user intent in natural-language grammar queries. The system must accurately identify what grammatical aspect a student is asking about from naturally-phrased grammar questions. Unlike simple word-based lookup, learners may phrase questions in various forms, and the system must be able to interpret these and map the questions to the grammatical concepts being queried. Additionally, the system must also be capable of producing accurate and useful feedback. The system must correctly identify and explain the grammatical features of words. Furthermore, the system must handle ambiguous forms where a single word form may correspond to multiple morphological analyses depending on context. This information should be delivered conversationally as accessible explanations that help learners understand these grammatical concepts.

1.3 Aims and Objectives

The aim of this project is to develop a conversational tool for Latin language learners that is there to help them with and provide answers to queries related to Latin grammar.

In order to achieve this aim, the following objectives have been identified:

- Construct a dataset of Latin words and their grammatical features by using existing Latin language datasets and expanding them.
- Design and generate a synthetic dataset for intent classification, consisting of grammar-related questions covering different grammatical questions. This will serve as training data for the intent classification model.
- Fine-tune a BERT-based model for intent classification to identify user grammatical queries.
- Develop a Response Generation System that, according to the intent classification, will carry out the necessary steps to construct the appropriate answer.
- Implement a conversational interface that accepts natural language input in English, so that users can interact with and submit grammar questions to the model, and receive the output in the form of a response.
- Evaluate the system's performance through quantitative measures and qualitative user testing with Latin learners.

2 Literature Review

This chapter presents the literature and research relevant to the development of the proposed Latin grammar chatbot. It examines the role of Artificial Intelligence in language learning, the development and use of chatbots, the available datasets and digital tools for Latin, BERT-based models, and finally intent classification and its application for query understanding. Together, these areas provide the foundation for the system proposed in this project.

2.1 AI in Language Learning

2.2 Chatbots

2.3 Datasets

Choosing the right dataset is fundamental to this project as it serves as the foundation for the model itself. The Universal Morphology (UniMorph) dataset (Sylak-Glassman et al., 2015) provides morphologically annotated word forms for multiple languages, including Latin. The Latin portion of UniMorph consists of entries containing a lemma, an inflected word form derived from that lemma, and a set of morphological features encoded using the UniMorph feature schema. These features indicate the different grammatical categories that represent the inflected word. The UniMorph dataset offers several advantages for this project. First, its standardized annotation format facilitates the representation of features for all words, making it consistent and easy to understand. Second, it covers a large amount of Latin words. Third, its machine-readable format makes it straightforward to integrate into computational systems. For these reasons this dataset was the main dataset chosen for this project. However, UniMorph has certain limitations relevant to this project. Most significantly, the dataset excludes specific features that are frequently requested by Latin learners seeking to understand inflectional patterns, particularly conjugation numbers for verbs and declension numbers for nouns. These gaps require strategies to expand and create a more complete dataset which are discussed in subsequent sections.

Other Latin resources include syntactically annotated corpora such as the PROIEL Treebank (Haug and Jøhndal, 2008) and the Perseus Latin Dependency Treebank (Bamman and Crane, 2011). Unlike UniMorph, which focuses on isolated words, these treebanks are made up of sentences from Latin texts with syntactic information. Both treebanks include texts by classical authors.

While morphological and syntactic datasets exist for Latin, there is an absence of datasets specifically designed for classifying Latin grammar questions. Most publicly available intent classification datasets focus on domains such as customer service, for example the dataset presented by Casanueva et al. (2020) that is made up of customer service questions in the banking domain. Therefore, a custom intent classification dataset of learner-style grammar questions must be created for this project.

2.4 Tools and Digital Resources for Latin

In addition to datasets, various computational tools and digital resources support Latin language learning and processing.

William Whitaker's Words is a morphological parsing program that analyses Latin words entered by users and returns grammatical information about inflected forms (Whitaker, 1993). The program's output includes the lemma, English meaning, and grammatical features such as case, number, gender, tense, mood, voice, person, conjugation number, and declension number. This output format resembles the structure of the UniMorph dataset, but Whitaker's Words often provides additional features absent from UniMorph, most notably the conjugation and declension numbers, which are significant for learners who want to know the set of inflectional patterns for a word. Whitaker's Words is a valuable resource for extending the grammatical dataset for this project. By automatically querying the program with word forms from UniMorph, and extracting additional feature information, a more comprehensive dataset can be constructed.

The Classical Language Toolkit (CLTK) is a Python library offering NLP tools specifically designed for classical languages, including Latin (Johnson et al., 2021). CLTK provides functionalities such as tokenization, lemmatization, part-of-speech tagging, named entity recognition, and text normalization. These tools are built with awareness of the specific characteristics of classical languages that other NLP libraries may not handle appropriately.

2.5 BERT and Latin BERT

BERT (Bidirectional Encoder Representations from Transformers), introduced by Devlin et al. (2019), is a language representation model that implements a Transformer encoder architecture that processes text bidirectionally by considering both left and right context simultaneously.

BERT is pre-trained on unlabeled data using two unsupervised tasks, a masked language model and next sentence prediction. This pre-training enables BERT to learn general language representations. The pre-trained BERT model can then be fine-tuned for downstream tasks, such as intent classification, by adding task-specific layers and training on labelled data for that task. This transfer learning technique is very effective, particularly for low-resource tasks, because the model makes use of knowledge from pre-training, rather than having to learn entirely from scratch (Devlin et al., 2019). For this project, fine-tuning BERT for the intent classification task on Latin grammar questions will enable the model to recognize certain patterns such as question structures and grammar-related terminology.

Recognizing the limitations of generic pre-trained models for Latin-specific tasks, Bamman and Burns (2020) developed Latin BERT, a BERT model trained specifically on Latin texts. Latin BERT was trained on a large corpus of Latin text from several sources, enabling it to learn Latin-specific language representations that generic multilingual BERT models might not capture effectively, which offers many improvements for Latin-specific NLP tasks. Bamman and Burns (2020) demonstrate the usefulness of Latin BERT with four case studies of different NLP tasks, showing Latin BERT outperforms other models.

2.6 Intent Classification

Intent classification is an important component of certain dialogue systems, enabling chatbots to understand the user's goals based on natural language input. It is commonly used in customer service chatbots, task-oriented dialogue systems, and question-answering systems. Intent classification involves categorizing user utterances according to their underlying goal. The dialogue system uses the classified intent to determine what action to take. Accurate intent classification is very important to chatbot effectiveness because misclassification leads to inappropriate responses.

Various approaches have been applied to intent classification as listed by Hoque et al. (2025) and Ran et al. (2025). Traditional intent classification approaches have used machine learning methods such as Support Vector Machines, logistic regression, and Naive Bayes. Deep learning-based approaches such as Convolutional Neural Networks and Recurrent Neural Networks have also been used for this task. More recently, the use of transformer-based models, such as BERT, for intent classification has achieved good results. For instance, Chen et al. (2019) propose a joint intent classification and slot filling model based on BERT, with results showing that it outperformed attention-based RNN models and slot-gated models on benchmark datasets.

This project adapts intent classification to the specific domain of Latin grammar by treating grammatical concepts as intents in a dialogue system. This adaptation reframes the intent classification task as identifying which grammatical aspect the learner is asking about.

3 Proposed Solution and Methodology

This section will discuss the current proposed solution to achieve the objectives of the project as defined in Section 1.3. The research carried out and discussed in the previous section will form a basis for this project, and the following subsections will discuss how the results produced by the literature can be used to solve the project's objectives.

3.1 System Overview

The overall proposed system can be described as a combination of the following components that will interact together:

- Intent classification module
- Response generation module
- Conversational interface

The main workflow of the system will be as follows:

1. User submits a natural-language grammar question in English.
2. Intent classification model identifies the grammatical category.
3. System retrieves the Latin grammar knowledge and generates a response.
4. Chatbot returns the response.

3.2 Dataset Creation

3.2.1 Latin Words Dataset

A dataset of Latin words and their grammatical features will be created by using the UniMorph dataset as the base dataset and using the Whitaker's Words program to fill in and add additional grammar information to the entries.

The UniMorph dataset was selected as it contains many entries of individual Latin words, along with their lemma and the grammatical features for how the word was inflected. Having a large collection of words and their corresponding grammatical features is needed for when it comes to constructing the intent classification dataset, which will contain questions about the grammar categories of specific words. While UniMorph is a good starting point for data, it has some limitations. The first limitation is that it does not include any information about the declension numbers of Latin nouns, nor does it include the conjugation number of Latin verbs. These features are important for Latin learners as they reveal the set of patterns on how the word changes form according to different grammatical categories. Another limitation of UniMorph is that for certain parts of speech, some grammatical categories are left out. For example, gender information is not included with nouns, even though Latin nouns do have a grammatical gender.

To address these limitations, Whitaker's Words will be used to fill in the gaps. This program was chosen since the output that it gives is already similar to the format of the UniMorph entries, and it includes the additional information that is missing. Since it is a command line program, it can be easily interacted with and controlled programmatically so that the extraction of the needed information can be automated.

The result will be a comma separated file of many entries of Latin word forms along with their grammatical features.

3.2.2 Intent Classification Dataset

A structured dataset of grammar-related question templates and intents will be created to be the training data for the intent classification model discussed in the following subsection.

The intent categories will be what grammar feature the user is asking about for a specific word. For example, if the user is asking about what tense a word is in, that question will be labelled as a tense query. The chosen intent categories are based on the different grammatical features covered in the dataset described in Section 3.2.1. These categories should cover common questions learners may have about specific Latin words. Another category will be defined to cover out of scope inputs.

The dataset will be created by first creating example question templates for each intent category. The templates will include placeholders so that when generating the questions, the appropriate Latin words will be inserted in their place. The templates will be created by manually writing different example questions, as well as making use of large language models to help generate variations of different ways of how the same question can be phrased. The data augmentation techniques that will be used to increase the number of different question templates include paraphrasing of questions, both by replacing synonyms and varying the structure of the sentences.

These templates will then be used to automatically generate many variations of questions to train the model, by taking a Latin word from the dataset described in the previous section and inserting it in the template. Along with each generated question, the corresponding intent will also be recorded in the dataset. This will create question and intent pairs for the classification task.

3.3 Intent Classification Model

An intent classification model will be created by fine-tuning a BERT-based model. This model will be used to understand the user queries. A BERT model was selected as it was discovered to be a popular approach that produced some of the best results in comparison to other methods (Larson et al., 2019; Hoque et al., 2025). The model will be fine-tuned using the dataset described in the previous section, which will be split into a train, validation, and test set. This system will focus on classifying the questions given as input into one of the grammatical intent categories. These grammatical intents will identify what specific grammar category the user was asking about, for example asking about the tense of the word. To handle ambiguous questions or unsupported queries, an additional intent category will be defined so that the model can classify questions that are not understood or out of scope as such, similarly to the approach described by Larson et al. (2019). This will ensure that this type of input is handled and the chatbot will still be able to return an appropriate response to the user, indicating that the question was not understood.

3.4 Response Generation

Once the intent of the question is known, the chatbot must produce explanations to return as an answer. The approach chosen for the response generation system is a template-based and retrieval-based approach. Various answer templates will be designed for each grammar intent category. These templates will contain placeholders for the insertion of the correct information. The grammatical information will be incorporated into the response by retrieving the required information for the word that the user is asking about from the dataset, and placing the information within the template. This retrieval-based approach will ensure better accuracy in the grammar information returned. The steps for producing a response are as follows:

1. Model identifies intent.
2. System extracts the Latin word from the sentence.
3. System looks up that word in the dataset.
4. System constructs a template response.

3.5 Conversational Interface Design

A conversational interface will be designed and implemented so that the user can ask questions to the chatbot and receive answers. This will be the user-facing component of the system, which will enable real-time interaction and reflects the project's goal of a conversational tool. The choice of interface will first be a command-line prototype for testing and then a web-based interface for the final system to enhance usability. The user interaction flow will consist of a query input mechanism for the user to input their question as text, feedback during processing as the user waits for the response, and a presentation format for the system's response.

4 Evaluation

This section outlines the proposed evaluation strategy for assessing the Latin chatbot's performance. The evaluation will include both quantitative and qualitative methodologies to provide a comprehensive assessment of the system's performance and accuracy, as well as its usability and usefulness for Latin students. Since the system incorporates multiple components, including dataset construction, intent classification, response generation, and user interaction, each component will be evaluated using the appropriate metrics and methods. The following subsections outline the planned evaluation methodologies.

4.1 Technical Evaluation (Quantitative)

A technical evaluation of the system will be conducted to assess the performance of the intent classification model and the operational performance of the chatbot.

4.1.1 Intent Classification Model Evaluation

The fine-tuned BERT model for intent classification will be evaluated on a held-out test set. Performance will be measured using standard classification metrics, these being accuracy, precision, recall, and F1 score, to evaluate how effectively the model identifies grammar-related intents.

4.1.2 System Performance Evaluation

The system will be tested to assess its reliability, examining how the system responds to ambiguous or malformed questions, and unknown words or rare inflections. This includes tracking error and fallback rates, such as how often the model fails to predict an intent with high confidence and returns a default error message. Additionally, response time measurements will be taken to evaluate how quickly the chatbot processes queries and returns explanations, to ensure the interaction feels conversational and real-time.

4.2 User Evaluation (Qualitative)

Qualitative evaluation will be carried out to determine the educational effectiveness and usability of the chatbot, and the clarity and correctness of its responses. This involves assessments by actual users.

4.2.1 Chatbot Evaluation

A manual review of the responses provided by the chatbot will be conducted to validate the quality of the generated responses, obtaining feedback from Latin instructors where possible. This will help validate the grammatical accuracy and appropriateness of explanations, as well as to see how it handles complex or ambiguous questions.

Additionally, in order to evaluate the perceived usability and usefulness of the system, Latin learners will interact with the system and feedback will be collected. Feedback will focus on the clarity of the explanations provided, the correctness of the grammatical information, the ease of communicating with the chatbot, whether they feel the chatbot supports their learning, and suggestions for improvement. This feedback will be used to identify issues in the conversational interface, and also evaluating whether the explanations provided are clear and understandable for a student, rather than just being technically correct.

Finally, comparisons will be made with existing Latin learning tools, such as Whitaker's Words and the Perseus Latin Word Study Tool (Crane). This comparison will highlight the difference between the proposed chatbot and traditional tools, particularly its ability to interpret natural-language grammar questions and provide a conversational explanation.

4.2.2 Dataset Evaluation

Since dataset quality directly affects model accuracy and system performance, the constructed dataset will also be evaluated. The question templates for the synthetic question dataset will be reviewed to ensure that the dataset reflects real or plausible learner questions, that it covers a wide variety of ways a student might actually phrase a question, and for any potential ambiguities in the phrasing.

5 Conclusion

This project aims to address the lack of a conversational assistant that can understand Latin grammar-related questions and respond interactively. The goal is to design a system that bridges the gap between classical language education and modern AI tools. Previous work highlighted that artificial intelligence is effective in language learning, particularly through conversational systems that provide interactive help, and existing tools such as UniMorph, Whitaker's Words, and BERT provide a basis for the proposed solution. The proposed approach consists of constructing a dataset, developing a grammar-focused intent classification model, and integrating these components into a conversational interface capable of providing good explanations, to produce a tool that supports Latin learners in understanding grammar. The next stages of the project will involve implementing these components and evaluating their effectiveness.

References

David Bamman and Patrick J. Burns. 2020. Latin bert: A contextual language model for classical philology. *Preprint*, arXiv:2009.10053.

David Bamman and Gregory Crane. 2011. The ancient greek and latin dependency treebanks. In *Language Technology for Cultural Heritage*, pages 79–98, Berlin, Heidelberg. Springer Berlin Heidelberg.

Iñigo Casanueva, Tadas Temčinas, Daniela Gerz, Matthew Henderson, and Ivan Vulić. 2020. Efficient intent detection with dual sentence encoders. In *Proceedings of the 2nd Workshop on Natural Language Processing for Conversational AI*, pages 38–45, Online. Association for Computational Linguistics.

Qian Chen, Zhu Zhuo, and Wen Wang. 2019. Bert for joint intent classification and slot filling. *Preprint*, arXiv:1902.10909.

Gregory R. Crane. Perseus digital library.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Dag T. T. Haug and Marius L. Jøhndal. 2008. Creating a parallel treebank of the old indo-european bible translations. In *Proceedings of the Second Workshop on Language Technology for Cultural Heritage Data (LaTeCH 2008)*, pages 27–34.

Fardinul Hoque, Rezaul Karim, Ratnadip Kuri, Kazi Nazmul Hasan, and ARM Mahamudul Hasan Rana. 2025. Intent classification in chatbots with human-ai collaborated large-scale dataset. In *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, pages 1–7. IEEE.

Kyle P. Johnson, Patrick J. Burns, John Stewart, Todd Cook, Clément Besnier, and William J. B. Mattingly. 2021. The Classical Language Toolkit: An NLP framework for pre-modern languages. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing: System Demonstrations*, pages 20–29, Online. Association for Computational Linguistics.

Stefan Larson, Anish Mahendran, Joseph J. Peper, Christopher Clarke, Andrew Lee, Parker Hill, Jonathan K. Kummerfeld, Kevin Leach, Michael A. Laurenzano, Lingjia Tang, and Jason Mars. 2019. An evaluation dataset for intent classification and out-of-scope prediction. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 1311–1316, Hong Kong, China. Association for Computational Linguistics.

Zhou Ran, Yuhan Chen, Quanli Jiang, and Kunpeng E. 2025. Intent recognition in dialogue systems. In *Proceedings of the 4th Asia-Pacific Artificial Intelligence and Big Data Forum*, AIBDF '24, page 165–173, New York, NY, USA. Association for Computing Machinery.

John Sylak-Glassman, Christo Kirov, David Yarowsky, and Roger Que. 2015. A language-independent feature schema for inflectional morphology. In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 674–680, Beijing, China. Association for Computational Linguistics.

William Whitaker. 1993. William Whitaker's Words. Available at https://mk270.github.io/whitakers-words/.

A Work Plan - Gantt Chart

Gantt Chart